MILITARY SPECIFICATION

RADAR IDENTIFICATION TRANSPONDER SET AN/APX-71A

This Military Specification has been approved by the
Naval Air Systems Command Department of the Navy.

1. SCOPE

1.1 Scope - The AN/APX-71A equipment described herein is an airborne transponder for use in conjunction with a system of electronic identification. The equipment shall provide automatic radar identification of an airborne vehicle so equipped to all suitably equipped challenging aircraft, surface ships, and ground forces operating within the operational range of the system.

1.2 Classification - Equipment covered by this specification shall consist of the following items:

Unit	Type Designation
Coder-Receiver Transmitter	KY-616 /APX-71A
Base, Shock Mount Electrical Equipment	MT-4518/APX-71A

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.5.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5400 Electronic Equipment, Aircraft, General Specification for

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MIL-T-5422	Testing Environmental, Aircraft Electronic Equipment
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-17555	Preparation for Delivery of Electronic Electrical and Electro-Mechanical Equipment and Associated Repair Parts
MIL-T-18303	Test Procedures; Preproduction and Inspection for Aircraft Electronic Equipment, Format for

STANDARDS

Military

MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-STD-481	Configuration Control-Engineering Changes, Deviations and Waivers (Short-Form)
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
Other	Order 1010.51, Department of Transportation U.S. National Standard for IFF MARK X(SIF) Air Traffic Control Radar Beacon (ATCRBS) Characteristics

DRAWINGS

Naval Air Systems Command

DL 386AS100	Data List for Transponder AN/APX-71A and all drawings listed therein

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WR-98 15 December 1965	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of

PUBLICATIONS

Military

ANA Bulletin No. 400W	Electronic Equipment; Aircraft and Guided Missiles, Applicable Documents

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials -

3.2.1 General Requirements - The parts and materials specified in the drawings and related documents for use in construction of this equipment were selected to ensure the fulfillment of all requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change, and the cost associated therewith, upon the contractor. Approval for deviations must be requested from the procuring activity by the contractor and granted by the procuring activity prior to submission of equipments for preproduction tests. Approval of parts and material deviations shall not be construed by the contractor to constitute a waiver of the performance requirements or any other requirements of this specification.

3.2.1.1 All Parts and Materials - Requirements shall conform to MIL-E-5400.

3.2.1.2 Nonreparable Subassemblies - As outlined in MIL-E-5400, non-reparable subassemblies, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonreparable subassemblies must be reliable.

3.2.1.3 Workmanship - Equipments manufactured shall meet Requirement 5 (Soldering) and Requirement 9 (Workmanship) of MIL-STD-454.

3.2.2 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices, shall be obtained as specified in MIL-E-5400. Microelectronic devices shall be approved in accordance with WR-98.

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3.2.3 Microelectronic Modular Assemblies - When used, microelectronic modular assemblies shall meet the requirements of WR-98.

3.2.4 Supplemental Requirements - In the event that the contractor proposes to alter the equipment design delineated by this specification and applicable drawings, the following supplemental requirements for the selection of parts and materials are applicable.

3.2.4.1 Basic Conformance - Parts and materials selected for use in the equipment shall conform to the requirements of MIL-E-5400. Parts shall be selected from those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform adequately in the intended applications and environments specified herein.

3.2.4.2 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400. The drawings listed in DL 386AS100 shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.3 Design and Construction - The equipment shall be manufactured in accordance with the drawings and subsidiary specifications listed in DL 386AS100 and related lists, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this and related specifications. In any case it is a condition of final acceptance that the equipment manufactured meet all the performance requirements of this specification, and such acceptance test documents as may be applicable. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be specifically approved by the procuring activity prior to their incorporation. Engineering changes, waivers and deviations shall be in accordance with MIL-STD-481.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 9.75 pounds excluding the shock mount, nor exceed 10.5 pounds including the mount specified in 3.5.

3.3.2 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall conform to the requirements of MIL-STD-461 within the limits established in 3.3.12.

3.3.3 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative
Input power voltage	27.5 ± 0.5 d-c
Interrogation rate	500 interrogations/sec when not otherwise specified or required

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3.3.4 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.3.4.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of MIL-E-5400.

3.3.4.2 Temperature - The equipment shall operate satisfactorily when subjected to the temperature requirements of Table I, Class 2 of MIL-E-5400.

3.3.4.3 Altitude - The equipment shall operate satisfactorily when subjected to the altitude requirements of Table I, Class 2, Column VIII of MIL-E-5400.

3.3.4.4 Temperature-Altitude Combination - The equipment shall operate satisfactorily when subjected to the requirements of Table I, Class 2, Column IV only.

3.3.4.5 Shock - The equipment shall meet the shock requirements of MIL-E-5400 except that impact shock shall be 20g.

3.3.4.6 Salt Spray - The equipment shall withstand exposure to salt spray in accordance with Procedure I of MIL-E-5272.

3.3.4.7 Pressurization - The equipment shall meet the pressurization requirements of MIL-E-5400.

3.3.5 Antenna Mismatch - Anticipated service conditions of impedance mismatch shall include all values of load impedance that present a voltage standing wave ratio (SWR) up to and including 3.5 decibels (DB) (1.497:1 SWR) of any phase relationship at the antenna terminal of the receiver-transmitter unit.

3.3.6 Transmitter Duty Cycle - The transmitter shall be capable of continuous operation at 1.0 percent duty cycle. For mean-time-between-failure (MTBF) and life-test considerations, operations shall be at 0.5 percent duty-cycle for 50 minutes and 1.0 percent for 10 minutes of each hour. For preproduction and acceptance testing, normal duty-cycle shall be 0.5 percent and limit conditions shall be 0.1 percent and 1.0 percent duty-cycles.

3.3.6.1 Duty Cycle - Percent duty cycle is defined by the following equation:

$$DC = \frac{(PW) \times (RC) \times (NP)}{10^4}$$

where:

DC is percent duty cycle
PW is pulse width in microseconds
RC is reply count in number of counts
NP is number of pulses

3.3.7 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 2 minutes under either standard or extreme service conditions.

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3.3.8 Input Electrical Power -

3.3.8.1 Operating Power - The equipment shall give specified performance when energized from the following power source. The power required shall not exceed the specified amounts.

3.3.8.1.1 D-C Input Voltage - 17 to 32 volts.

3.3.8.1.2 The input power shall not exceed 50 watts over the input voltage range of 17 to 32 volts d-c.

3.3.8.2 Degraded Performance - Degraded performance will be permitted for voltage transients of time duration not exceeding 0.5 second during normal AC or DC electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.9 Cooling - The equipment shall not require either an internal blower or external cooling air.

3.3.10 Interrogation Characteristics - The equipment shall accept interrogation signals with characteristics in accordance with tables I and II.

Table I. Interrogation Mode Spacing

Mode	Interrogation Characteristic
2	2 pulses spaced 5 ± 0.2 USEC
3	2 pulses spaced 8 ± 0.2 USEC

Table II. Input Pulse Characteristics

Mode	Interrogation Code	
2, 3	Pulse Width	0.8 ± 0.1 USEC
	Rise Time	0.05 to 0.1 USEC
	Decay Time	0.05 to 0.2 USEC
	Amplitude Variation	P3 = P1 ± 1 DB

3.3.11 Input Protection - The equipment shall not be damaged by external signals of any frequency or type of modulation of up to 10 volts peak amplitude and average power of -20 DBM applied directly to the antenna terminal.

3.3.12 Radio Frequency Interference Control - The equipment shall meet the requirements of test CS01, CS03, CS04, and RS03. With the transponder being interrogated in either mode 2 or mode 3 it shall meet the requirements of CE03, CE06, CS02, and RE02 as modified below (3.3.12.1 - 3.3.12.5).

3.3.12.1 Test Method CE03 - Conducted Emission, Power Line Test: The frequency range shall be 150 KHz to 25 MHz. The broadband (BB) limits shall be increased by 31 dB from 2 to 25 MHz; from 0.15 to 2.0 MHz increase the revised level by 9 dB per octave as the frequency decreased from 2 to 0.15 MHz.

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3.3.12.2 Narrowband (NB) Limits: The limits shall be increased by 14 dB from 5 to 25 MHz. From 0.150 to 5 MHz increase the revised level by 5.5 dB per octave as the frequency decreases from 5 to 0.15 MHz.

3.3.12.3 Test Method CE06 - Antenna Conducted Spurious Emanations Test: The BB limits shall be increased by 6 dB in the frequency range from 10 KHz to 12.4 GHz.

3.3.12.4 Test Method CS02 - Conducted Susceptibility, Power Lines Test: In the frequency range of 50 KHz to 400 MHz the specified 1.0 V rms test level shall be reduced to 0.1 V rms.

3.3.12.5 Test Method RE02 - Radiated Emissions, Electric Field Test: In the frequency range of 10 KHz to 10 GHz (CW) and 150 KHz to 1.0 GHz (BB) increase the BB limits by 3 dB per octave from 200 MHz to 1.0 GHz. Increase the CW limits by 8.5 dB per octave from 300 MHz to 1.0 GHz and increase the limits by 5 dB per octave as the frequency is decreased from 10 to 1.0 GHz. The transmitter fundamental frequency (1090 Mc) shall not be included in the tests of 3.3.12.5.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall receive coded interrogations (3.3.10), shall decode these interrogations, and in response, shall transmit proper pulse replies.

3.4.1.1 Input Voltage Range - The equipment shall give full specified performance when operated with steady-state input voltage from 21 to 32 volts d-c. Full specified performance shall be restored when the input voltage returns to normal.

3.4.1.1.1 The equipment shall operate with the input voltage in the range of 17 to 21 volts d-c, with reduced performance as detailed in 3.4.3.5 and 3.4.7.3.1.

3.4.1.2 Interference Characteristics - The equipment shall operate in the presence of the following types of interfering signals with minimum degradation:

3.4.1.2.1 Continuous Wave and Amplitude Modulated Continuous Wave Interference - The equipment shall operate in the presence of interfering signals consisting of continuous wave (CW) and amplitude-modulated continuous wave (AMCW) and shall meet the following requirements. Measurements shall be made to assure compliance throughout the required range of interrogating signal levels.

(1) If the interrogating signal to which the equipment should reply is between 3 and 50 db above the normal triggering level, an interfering signal 10 db below the interrogating signal shall not reduce the number of valid replies more than 50 percent. At each level tested the interfering signal frequency shall be adjusted for maximum countdown of replies.

(2) The AMCW signal shall be simulated using a signal generator modulated 30 percent by a 1000 Hertz (Hz) sine wave.

(3) If the interrogating signal is between 20 and 50 db above the normal triggering level, an interfering signal at a level 3 db down from the interrogating signal shall not reduce the number of replies by more

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than 50 percent, except when the interfering signal combines with the interrogating signal in such a way that the amplitude modulation of the resultant signal is inverted or eliminated.

3.4.1.2.2 Pulse Interference - The receiver shall operate in the presence of interfering signals consisting of randomly spaced pulses and regularly spaced pulses and shall meet the following requirements:

(1) If the interrogating signal to which the equipment should reply is between 3 and 10 db above the normal triggering level, an interfering signal 10 db down from the interrogating signal shall not reduce the number of replies by more than 50 percent.

(2) If the interrogating signal is between 20 and 50 db above the normal triggering level, an interfering signal 3 db down from the interrogating signal shall not reduce the number of replies by more than 50 percent.

(3) If the interfering signal consists of randomly-spaced pulses only, and the interrogating signal is a Mode 2 or 3 interrogation that is 10 db above the interfering signal, the equipment shall reply to at least 90 percent of the interrogations where the randomly spaced pulses are not recognized as normal or side-lobe interrogations.

3.4.2 Receiver Detailed Performance Requirements - The receiver shall be of the tuned radio-frequency crystal video type incorporating at least three stages of radio frequency amplification prior to the detector.

3.4.2.1 Receiver Frequency - The receiver shall operate at a fixed frequency of 1030 MHz.

3.4.2.2 Receiver Frequency Stability - Under all combinations of service conditions specified herein, the frequency drift from that measured under conditions of 3.3.3 shall not exceed ±1.5 MHz.

3.4.2.3 Receiver Noise Figure - The receiver noise figure shall not exceed 13 db.

3.4.2.4 Receiver Dynamic Range - Performance requirements specified in 3.4.2 and subparagraphs shall be met with interrogation signals from minimum triggering level to at least 50 db above that level unless otherwise specified.

3.4.2.5 Receiver Bandwidth - Overall receiver bandwidth shall be in accordance with table III. The bandpass shall be essentially gaussian and shall accommodate interrogation signal frequencies of 1030 ± 2.5 MHz with not more than 3 db variation in sensitivity.

Table III. Overall Receiver Bandwidth

-6 db points	greater than 7.0 MHz
-60 db points	less than ±25 MHz

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3.4.2.6 Receiver RF Rejection - The CW and pulse responses shall be 60 db less than that measured at 1030 MHz for all frequencies less than 1005 MHz and greater than 1055 MHz.

3.4.3 Sensitivity -

3.4.3.1 Desensitization - Upon receipt of any pulse more than 0.7 USEC in duration, the receiver shall be desensitized by an amount that is within at least 9 db of the amplitude of the desensitizing pulse, but shall at no time exceed the amplitude of the desensitizing pulse, with the exception of possible over-shoot during the first USEC following the desensitizing pulse.

3.4.3.1.1 Single pulses of duration less than 0.7 USEC are not required to cause the specified desensitization, and should not cause desensitization greater than permitted by 3.4.3.1.

3.4.3.2 Recovery - The receiver shall remain desensitized for at least 2.5 microseconds following reception of a desensitizing pulse of signal strength up to 50 db above the minimum trigger level (MTL). The receiver sensitivity shall then recover linearly at a rate not exceeding 4.5 db per microsecond within 15 microseconds to within 3 db of MTL.

3.4.3.3 Minimum Triggering Level - Minimum triggering level is that level of signal required for 90 percent replies. The MTL of the equipment at the receiver antenna jack, when measured with nominal interrogation modes and at a frequency of 1030 ± 0.2 MHz under standard conditions (3.3.3) shall be at least -85 decibels referred to 1 volt (DBV) (-72 decibels referred to 1 milliwatt into 50 ohms (DBM) for all modes). The total variation of MTL between Modes 2 and 3 shall not exceed 1.0 db when measured as above. Also, the MTL of the equipment on Modes 2 and 3 shall not change by more than ±1.0 db when the interrogation pulse width is varied over the tolerance specified in 3.3.10 for each mode.

3.4.3.3.1 MTL Adjustment - The MTL shall be adjustible between -80 DBV and -90 DBV. It shall be adjusted to the level specified in 3.4.3.3 prior to delivery.

3.4.3.4 Triggering Level Drift - The triggering level, when set at -85 DBV (-72 DBM) under standard conditions (3.3.3) shall not vary by more than ±3 db over all combinations of service conditions.

3.4.3.5 Reduced Triggering Level Performance - The MTL as measured under conditions of 3.3.3 may be reduced no more than 6.0 db when the input voltage is reduced to 17.0 volts d-c.

3.4.3.6 Random Triggering Rate - The equipment shall not transmit more than five pulse trains per second averaged over 1 minute under the following conditions:

(1) The receiver antenna receptacle is terminated in 51-ohm resistive load.

(2) No external signal is present.

(3) Triggering level (sensitivity) control is set for an MTL of at least -85 DBV (-72 DBM) as required in 3.4.3.3.

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3.4.4 Overall Performance Requirements -

3.4.4.1 Automatic Overload Control - The automatic overload control (AOC) shall operate, by means of trigger level-reduction, to perform the following functions:

3.4.4.1.1 Reply Duty-Cycle Control - The reply duty-cycle shall be controlled to protect the transmitter. The AOC action shall be equally effective in all modes. No more than 3 db reduction in sensitivity shall occur at 0.9 percent duty-cycle. Maximum duty-cycle shall not exceed 1.1 percent when interrogation signals are at 50 db above minimum triggering level.

3.4.4.1.2 Reply Rate Limiting - This control shall limit the number of generated reply trains in Modes 2 and 3, without regard to the number of pulses in each reply. Trigger level reduction shall be gradual to prevent weaker interrogations from denying transponder service to stronger interrogations as a result of over-interrogation. Maximum reply rate shall not exceed 120 percent of the rate at which reduction first begins if the interrogation signal is increased to 50 db above MTL. Maximum reply rate shall be adjustable from 500 to 2000 replies per second, with interrogation signals 3 db above MTL, and shall be set to 1200 replies per second prior to delivery.

3.4.4.1.3 Automatic Overload Control Stability - The absolute limits stated in 3.4.4.1.1 shall apply under all service conditions. The rates established by adjustment in 3.4.4.1.2 under conditions of 3.3.3 shall hold within $\pm$20 percent under all service conditions.

3.4.4.1.4 Reply Rate Limiting Action and Recovery Time - If the interrogation rate is suddenly increased from 1000 to 2500 pulse pairs per second when the threshold established by the control described in 3.4.4.1.2 has been adjusted to 2000 replies per second, the sensitivity of the receiver shall be reduced by 20 db, for interrogation pairs 20 db above MTL, within 0.2 seconds. Upon decreasing the interrogation rate to 1000 pulse pairs per second the receiver shall return to within 3 db of normal sensitivity within 0.025 seconds.

3.4.4.2 Suppression Characteristics -

3.4.4.2.1 Suppression During Transmission of Replies in Modes 2 and 3 - Decoding of Modes 2 and 3 shall be suppressed during transmission of replies. Suppression shall continue for at least 75 USEC after the second transmitted framing pulse, but no longer than 125 USEC.

3.4.4.3 Echo Rejection - The transponder shall contain echo rejection circuitry designed to permit normal operation in the presence of echoes of signals in space and be compatible with the requirements specified in 3.4.4.4.

3.4.4.4 Interrogation Side Lobe Suppression (ISLS) -

3.4.4.4.1 ISLS Control Pulse Characteristics - The ISLS control pulse will have characteristics as listed in table IV.

3.4.4.4.2 Conditions for ISLS - The decoding of Modes 2 and 3 shall be suppressed (not less than 99 percent efficiency for interrogations 3 db or more above MTL) when the received amplitude of the ISLS control pulse is equal to or in excess of the received amplitude of the first interrogation pulse and spaced 2 $\pm$

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0.15 USEC from it. The suppression of the decoder shall be for a period of 35 ± 10 USEC and shall be capable of being reinitiated for the full duration within 2 USEC after the end of the suppression period.

Table IV. ISLS Control Pulse Characteristics

Characteristics	Value
Pulse Width	0.8 ± 0.1 USEC for Modes 2 and 3 interrogations
Pulse Position	2 ± 0.15 USEC after the first pulse of a Mode 2 and 3 interrogations
Rise Time	Between 0.05 and 0.1 USEC
Decay Time	Between 0.05 and 0.2 USEC
Frequency	Within ±0.2 megahertz (MHz) of the frequency of the interrogation pulses (1030 MHz nominal)

3.4.4.4.3 ISLS Conditions for Non-Suppression - The suppression of the decoder shall not take place when the received amplitude of the first interrogation pulse is 9 db or more in excess of the received amplitude of the ISLS control pulse, or when the spacing of the first interrogation pulse and ISLS is more than 2.6 USEC.

3.4.4.4.4 ISLS Gray Region - If the level of the interrogation pulse is greater than the level of the suppression pulse but greater by less than 9 db, suppression may or may not take place.

3.4.4.4.5 ISLS Dynamic Range - The ISLS suppression and non-suppression characteristics shall apply over a received signal amplitude range from 3 db above MTL to greater than 50 db above that level.

3.4.4.5 Pulse Width Discrimination - Signals of received amplitude between MTL and 6 db above this level, and of a duration less than 0.3 USEC on Modes 2 and 3 at the antenna input, shall not cause the transponder to initiate more than 10 percent reply or suppression action. With the exception of pulses with amplitude variations approximating a normal interrogation or suppression pulse pair condition, any pulse or pair of pulses each of a duration more than 1.5 USEC shall not cause the transponder to initiate reply or suppression action.

3.4.5 Decoder Detailed Performance Requirements - The equipment shall incorporate decoding circuits for the purpose of distinguishing and causing replies to the modes of interrogation described in 3.3.10. In addition, interrogations made up of pulses of 0.6 to 1.2 USEC width with rise times of 0.2 USEC and decay time of 0.3 USEC, shall be decoded and replies generated with no degradation in performance, except that the decoder tolerance may be proportional to pulse width. This circuitry shall be such that the reply transmitted by the equipment shall be the same regardless of the mode of interrogation.

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3.4.5.1 Decoder Tolerance, Mode 2 - Interrogation pulses spaced 4.8 to 5.2 USEC shall cause the equipment to generate replies. Pulses spaced less than 4.2 or greater than 5.8 USEC shall not cause such replies.

3.4.5.2 Decoder Tolerance, Mode 3 - Interrogation pulses spaced 7.8 to 8.2 USEC shall cause the equipment to generate replies. Pulses spaced less than 7.2 or greater than 8.8 USEC shall not cause replies.

3.4.5.3 Mode Selection - Mode 2, Mode 3, or both modes of interrogation shall be selectable by means of an appropriate switch located on the front of the unit.

3.4.5.4 Transponder Delay - The transponder delay between the second pulse of a Mode 2 or 3 interrogation received at the antenna terminal and the first reply pulse delivered at the antenna terminal shall be between 2.5 and 3.5 USEC. The variation in delay between Modes 2 and 3 shall not exceed 0.2 USEC for nominally spaced interrogations.

3.4.5.5 Range Jitter - On both Modes 2 and 3 the total jitter of the reply pulse code group, with respect to P_3, shall not exceed 0.1 USEC for receiver input levels between 3 and 50 db above MTL.

3.4.6 Reply Characteristics - The transmitted replies as measured at the antenna terminal shall have the characteristics listed in table V.

Table V. Reply Pulse Characteristics

Characteristics	Value
Duration	0.45 ± 0.1 USEC
Rise time	0.05 to 0.1 USEC
Decay time	0.05 to 0.2 USEC
Amplitude Jitter	5 percent or less

3.4.6.1 Pulse Width Variation - The variation in width between the widest pulse and the narrowest pulse in the reply shall not exceed 0.1 microsecond.

3.4.6.2 Mode 2 and Mode 3 Replies - For each pulse appearing at the decoder output, a pulse train shall be transmitted containing from none to a maximum of 7 information pulses plus 2 framing pulses which are spaced 20.3 (±0.10) USEC. The information pulse spacing shall be in multiples of 1.45 USEC from the initial framing pulse with a maximum allowable deviation from this spacing of ±0.05 USEC. The use of the seventh information pulse position (10.15 ± 0.05 USEC from the framing pulse) is covered in 3.4.6.5. From the specified 6 information pulses, a total of 64 different codes are selectable.

3.4.6.3 Code Number Assignments - Every configuration of Mode 2 and 3 information pulses shall be assigned a code number.

3.4.6.4 Codes - Code numbers 00 through 77 shall designate the Mode 2 and 3 codes. The applicable six information pulses shall be divided into two groups of

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three pulses. The individual pulses of each group shall be spaced 2.9 USEC apart and numbered in order (1, 2, and 4). The position of the first pulse of each group, measured from the initial framing pulse, shall be as shown in table VI.

Table VI. Mode 2 and 3 Pulse Positions

Group	Time From First Framing Pulse
A	02.9 USEC
B	11.6 USEC

The appropriate code number shall be determined by adding the sum (1, 2, and 4) of the pulses present within each group. These sums shall then be combined in order (A, B,) to form a two-digit serial number for the particular code.

3.4.6.5 X-Pulse Codes - A pulse spaced 10.15 ± 0.05 USEC from the initial framing pulse shall be designated as the "X" pulse. This pulse position is normally unused. Provisions shall be made such that, when a single switch located on the front panel is actuated. Both replies (Modes 2, and 3) will include this pulse in addition to the normal framing and information pulses.

3.4.7 Transmitter Detailed Performance Requirements - The equipment shall respond to interrogations defined in 3.3.10 by transmitting pulse-train replies for each interrogation received, depending on the mode of interrogation and upon equipment control settings selected. The transmitter shall consist of a grid-modulated oscillator and an RF driven power amplifier.

3.4.7.1 Transmitter Frequency - The transmitter shall operate at a fixed frequency of 1090 MHz.

3.4.7.2 Transmitter Frequency Adjustment - Before the equipment is delivered by the contractor, the transmitter frequency shall be adjusted to 1090 (±0.5) MHz, when the input impedance presented to the equipment is such that a VSWR of not more than 1.1 to 1 exists.

3.4.7.2.1 Transmitter Frequency Stability - When the transmitter has been adjusted in accordance with 3.4.7.2 under standard conditions of 3.3.3 and then submitted to any combination of service conditions, the transmitter frequency shall remain at 1090 ± 3.0 MHz.

3.4.7.3 Transmitter Power Output - The transmitter shall deliver 25 ± 3 db above 1 watt to the antenna terminal of the equipment under all combinations of specified conditions. When the equipment is delivered, the transmitter power output shall be at least 25 db above 1 watt when measured under the conditions of 3.3.3 and at a 1 percent duty cycle.

3.4.7.3.1 Transmitter Power Output Variation with Input Voltage - The transmitter peak power measured under the nominal conditions of 3.3.3 shall not be reduced by more than 3.0 db when the input supply voltage is reduced to 17.0 volts d-c.

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3.4.7.3.2 Transmitter Power Output Variation Within a Pulse Train - The peak power of any pulse in a group of pulses specified in 3.4.6 shall not differ from the power of any other pulse in that group by more than 1.0 db.

3.5 Mounting MT-4518/APX-71A - Unless otherwise specified in the contract an isolation mount in accordance with 386AS150, and capable of meeting the vibration requirements of 3.3.4.1 shall be provided. The overall dimensions of the Receiver-Transmitter complete with the isolation mount, shall not exceed 6.7 inches wide by 4.7 inches high by 13.5 inches including sway space.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests

(2) Acceptance Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.4. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of MIL-T-5422 and interference tests in accordance with MIL-I-6181.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

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4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection of an equipment and of equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, and operational adequacy. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual, and special tests.

4.3.3 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.3.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or a design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

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(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.5 Reconditioning of Test Equipment - Equipment which has been subjected to sampling tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Presubmission Testing - No item part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the Government Inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The radar identification transponder set is intended for utilization in aerial targets for use in conjunction with a system of electronic identification and recognition. When properly installed in such an aerial target, the equipment shall provide automatic radar identification of that target to all suitably equipped, challenging aircraft, surface ships, and ground forces operating within the operational range of the system.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a

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substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts, and processes other than those specified in DL 386AS100 it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) The Drawings of DL 386AS100 - The drawings shall have precedence over other referenced specifications. Any deviation from the drawings shall be specifically approved by the procuring activity.

(4) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Serial number identification.

6.5 Associated Equipment - The transponder shall operate with the following associated equipment:

(1) Antenna Assembly AS/133-APX

(2) Antenna Assembly AT-741/A

Project No. 5841-N041

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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